

# Ridgeback Concentrator - Operating Manual

Doc No. RB-4410-OM-001 Rev 2 - Cymbal Copper Pty Ltd (fictional)

## 1. Purpose and scope

This manual describes normal operation, start-up, shutdown and abnormal situations for the Ridgeback Concentrator (Areas 21 to 61). It is written for control room operators, metallurgists and maintenance planners.

Design values are taken from the process data sheets and the process flow diagrams (RB-4410-PFD-000 to 003). Where this manual gives an operating target it is noted as such.

## 2. Plant overview

Run-of-mine ore is crushed in the primary gyratory crusher CR-2101 and conveyed to the coarse ore stockpile. The grinding circuit (SAG mill ML-3101, pebble crusher CR-3102, ball mill ML-3201 and cyclone cluster CY-3201) treats 2,688 t/h at 93 % availability (60,000 t/d).

Cyclone overflow at P80 150 um feeds rougher flotation (FC-4101 to FC-4107). Rougher concentrate is reground in ML-4201 to P80 38 um and cleaned in column FC-4301 to a 26 % Cu concentrate (50.6 t/h). Concentrate is thickened (TH-5101) and filtered (FP-5101). Tailings (2,637 t/h) are thickened in TH-6101 and pumped to the TSF.

### 3.21 Area 21 - Primary Crushing

Principal equipment in Area 21:

| Tag     | Equipment                | Key design data                                                                  |
|---------|--------------------------|----------------------------------------------------------------------------------|
| CR-2101 | Primary Gyratory Crusher | Type: Gyratory 60 x 113; Installed power: 1,200 kW; Design throughput: 5,500 t/h |

### 3.31 Area 31 - SAG Milling

Principal equipment in Area 31:

| Tag     | Equipment      | Key design data                                                                                                             |
|---------|----------------|-----------------------------------------------------------------------------------------------------------------------------|
| ML-3101 | SAG Mill       | Diameter x EGL: 12.2 x 6.7 (40 ft x 22 ft) m; Installed power: 22,000 kW; Drive: Gearless motor drive (GMD), variable speed |
| CR-3102 | Pebble Crusher | Type: Cone crusher, extra-coarse; Installed power: 750 kW; Design throughput: 484 t/h                                       |

### 3.32 Area 32 - Ball Milling & Classification

Principal equipment in Area 32:

| Tag      | Equipment                               | Key design data                                                                                                          |
|----------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| ML-3201  | Ball Mill                               | Diameter x EGL: 8.2 x 13.4 (27 ft x 44 ft) m; Installed power: 22,000 kW; Drive: Gearless motor drive (GMD), fixed speed |
| PP-3201A | Cyclone Feed Pump (A/B duty, C standby) | Service: Cyclone feed; Configuration: 2 duty + 1 standby (PP-3201A/B/C); Rated flow: 5,539 m3/h                          |
| CY-3201  | Cyclone Cluster                         | Cyclones: 16 x 660 mm (13 operating / 3 standby); Feed pressure: 100 - 130 kPa; Feed density: 65 % solids w/w            |

### 3.41 Area 41 - Rougher Flotation

Principal equipment in Area 41:

| Tag     | Equipment                                         | Key design data                                                                                      |
|---------|---------------------------------------------------|------------------------------------------------------------------------------------------------------|
| FC-4101 | Rougher Flotation Tank Cells (FC-4101 to FC-4107) | Cells: 7 x 300 m3 forced-air tank cells; Residence time: 32 min; Air rate: 0.8 - 1.6 m/s superficial |

### 3.43 Area 43 - Regrind & Cleaner Flotation

Principal equipment in Area 43:

| Tag     | Equipment                       | Key design data                                                                |
|---------|---------------------------------|--------------------------------------------------------------------------------|
| ML-4201 | Regrind Mill (vertical stirred) | Installed power: 3,000 kW; Product P80: 38 um; Media: Ceramic 3 - 5 mm         |
| FC-4301 | Cleaner Flotation Column        | Diameter x height: 4.5 x 12 m; Wash water: 35 m3/h; Concentrate grade: 26 % Cu |

### 3.45 Area 45 - Reagents (Xanthate, Frother, Lime)

Principal equipment in Area 45:

| Tag     | Equipment                | Key design data                                                            |
|---------|--------------------------|----------------------------------------------------------------------------|
| TK-4501 | PAX Xanthate Mixing Tank | Volume: 40 m3; Solution strength: 20 % w/v; Material: 316L stainless steel |

|         |                      |                                                                                               |
|---------|----------------------|-----------------------------------------------------------------------------------------------|
| TK-4511 | Frother Storage Tank | Volume: 30 m3; Reagent: MIBC (methyl isobutyl carbinol); Flash point: 41 degC                 |
| TK-4521 | Milk-of-Lime Tank    | Volume: 150 m3; Slurry strength: 20 % w/w Ca(OH)2; pH control: AIC-4101, rougher feed pH 10.5 |

### 3.51 Area 51 - Concentrate Thickening & Filtration

Principal equipment in Area 51:

| Tag      | Equipment                            | Key design data                                                                         |
|----------|--------------------------------------|-----------------------------------------------------------------------------------------|
| TH-5101  | Concentrate Thickener                | Type: High-rate; Diameter: 30 m; Underflow density: 65 % solids w/w                     |
| PP-5101A | Concentrate Thickener Underflow Pump | Rated flow: 210 m3/h; Slurry SG: 2.0; Total dynamic head: 28 m                          |
| FP-5101  | Concentrate Filter Press             | Type: Vertical plate pressure filter; Filtration area: 144 m2; Cake moisture: 8.5 % w/w |

### 3.61 Area 61 - Tailings Thickening & Pumping

Principal equipment in Area 61:

| Tag      | Equipment                                         | Key design data                                                               |
|----------|---------------------------------------------------|-------------------------------------------------------------------------------|
| TH-6101  | Tailings Thickener                                | Type: High-rate; Diameter: 60 m; Underflow density: 62 % solids w/w           |
| PP-6101A | Tailings Pumps (3 stages in series, PP-6101A/B/C) | Rated flow: 2,576 m3/h; Slurry SG: 1.652; Stages: 3 in series to TSF (6.2 km) |

## 4. Grinding circuit operating targets

| Variable                      | Target          | Control                  |
|-------------------------------|-----------------|--------------------------|
| SAG mill feed rate            | 2,688 t/h       | WIC-3101                 |
| SAG mill solids               | 75 % w/w        | FIC-3103 water ratio     |
| SAG mill power                | 18 - 21 MW      | Speed and feed rate      |
| Ball mill power               | 19 - 21 MW      | -                        |
| Cyclone feed density (design) | 65 % solids w/w | DT-3201 / dilution water |
| Cyclone feed pressure         | 100 - 130 kPa   | PT-3201                  |
| Cyclone overflow P80          | 150 um          | PSI analyser             |
| Circulating load              | 250 - 300 %     | Calculated               |

Operating the cyclones above the design feed density coarsens the overflow and reduces rougher recovery. Operators should reduce feed rate before exceeding the target.

## 5. SAG mill start-up

- Confirm lube and hydrostatic lift system running, PT-3104 above 90 bar(g) (start permissive SIF-3102).
- Confirm TT-3101/TT-3102 healthy and below alarm (TAH 70 degC).
- Check for frozen (locked) charge after a stop longer than 4 hours: inch the mill using GMD inching mode and observe the charge.
- Start the mill at 70 % critical speed, then introduce feed at 50 % of target over 15 minutes.
- Start cyclone feed pumps PP-3201A and B before mill discharge reaches the hopper.

## 6. SAG mill shutdown and relining

For a planned stop, run the mill empty (grind-out) for 10 to 15 minutes to reduce charge weight. Isolate the GMD at the 33 kV feeder and apply personal locks per site standard RB-4410-SG-005 (mill isolation).

Relining uses the mill relining machine; the liner handler must not be used while the mill is on inching drive. Typical SAG reline duration is 72 to 96 hours.

## 7. Abnormal situations - grinding

| Symptom                      | Likely cause                                 | Response                                                |
|------------------------------|----------------------------------------------|---------------------------------------------------------|
| SAG bearing temperature high | Lube oil flow loss, cooler fouling           | TAH 70 degC alarm; SIF-3101 trips mill at TAHH          |
| Locked charge                | Stop longer than 4 h with fine ore           | Inch mill; never start at full speed with frozen charge |
| Cyclone roping               | Feed density too high or pump under capacity | Add dilution water, check pump speed                    |
| Hopper overflow / sanding    | Cyclone feed pump under capacity             | Reduce fresh feed; start standby pump                   |

## 8. Reagent handling

PAX (potassium amyl xanthate) is delivered as pellets and mixed to 20 % w/v in TK-4501. Xanthate decomposes in contact with moisture, heat or acid and releases carbon disulfide (CS2), which is highly flammable (LEL 1.3 %, auto-ignition 90 degC) and toxic.

- Keep the mixing room extraction fan FN-4501 running whenever PAX is present.
- Do not mix PAX with water above 40 degC.
- Use non-sparking tools in the mixing room.
- Dosing: PAX 25 g/t, frother 20 g/t, lime to pH 10.5.

Frother is stored in TK-4511 (MIBC, flash point 41 degC). The storage area is classified Zone 2.

Quicklime is slaked in SL-4521; the reaction is strongly exothermic. Wear face shields and chemical gloves when handling lime slurry.

## 9. Flotation operation

| Variable                | Target  |
|-------------------------|---------|
| Rougher feed pH         | 10.5    |
| Rougher residence time  | 32 min  |
| Regrind P80             | 38 um   |
| Final concentrate grade | 26 % Cu |
| Cu recovery             | 89 %    |
| Au recovery             | 70 %    |

## 10. Concentrate thickening and filtration

Maintain TH-5101 underflow at 65 % solids. Filter cake moisture must stay below the transportable moisture limit (TML 9.8 %); target 8.5 %.

## 11. Tailings pumping

Three pumps in series (PP-6101A/B/C) deliver 2,576 m3/h to the TSF at 62 % solids. Stage-3 discharge is normally 34 bar(g). PT-6103 PAHH at 38 bar(g) trips all pumps via SIF-6101 (SIL 2).

After a trip, flush the line with water before restart to prevent sanding. Report any pipeline pressure excursion to the Engineer of Record for the TSF.

## 12. Safety instrumented functions (summary)

| Tag      | Service                                             | Alarm / trip                                       | Function                           |
|----------|-----------------------------------------------------|----------------------------------------------------|------------------------------------|
| TT-3101  | SAG mill drive-end trunnion bearing temperature     | TAH 70 degC / TAHH 75 degC                         | SIF-3101 (SIL 1) stop SAG mill     |
| TT-3102  | SAG mill non-drive-end trunnion bearing temperature | TAH 70 degC / TAHH 75 degC                         | SIF-3101 (SIL 1) stop SAG mill     |
| PT-3104  | SAG hydrostatic lift oil pressure                   | PAL 90 bar(g) / PALL 70 bar(g)                     | SIF-3102 inhibit mill start / stop |
| PSV-3105 | SAG HP lift pump discharge relief                   | Set pressure 160 bar(g) (system design 170 bar(g)) | Mechanical relief                  |
| PT-6103  | Tailings stage-3 discharge pressure                 | PAH 36 bar(g) / PAHH 38 bar(g)                     | SIF-6101 (SIL 2) trip PP-6101A/B/C |

The cause & effect matrix RB-4410-STD-CE-001 is the controlling document for SIS logic.